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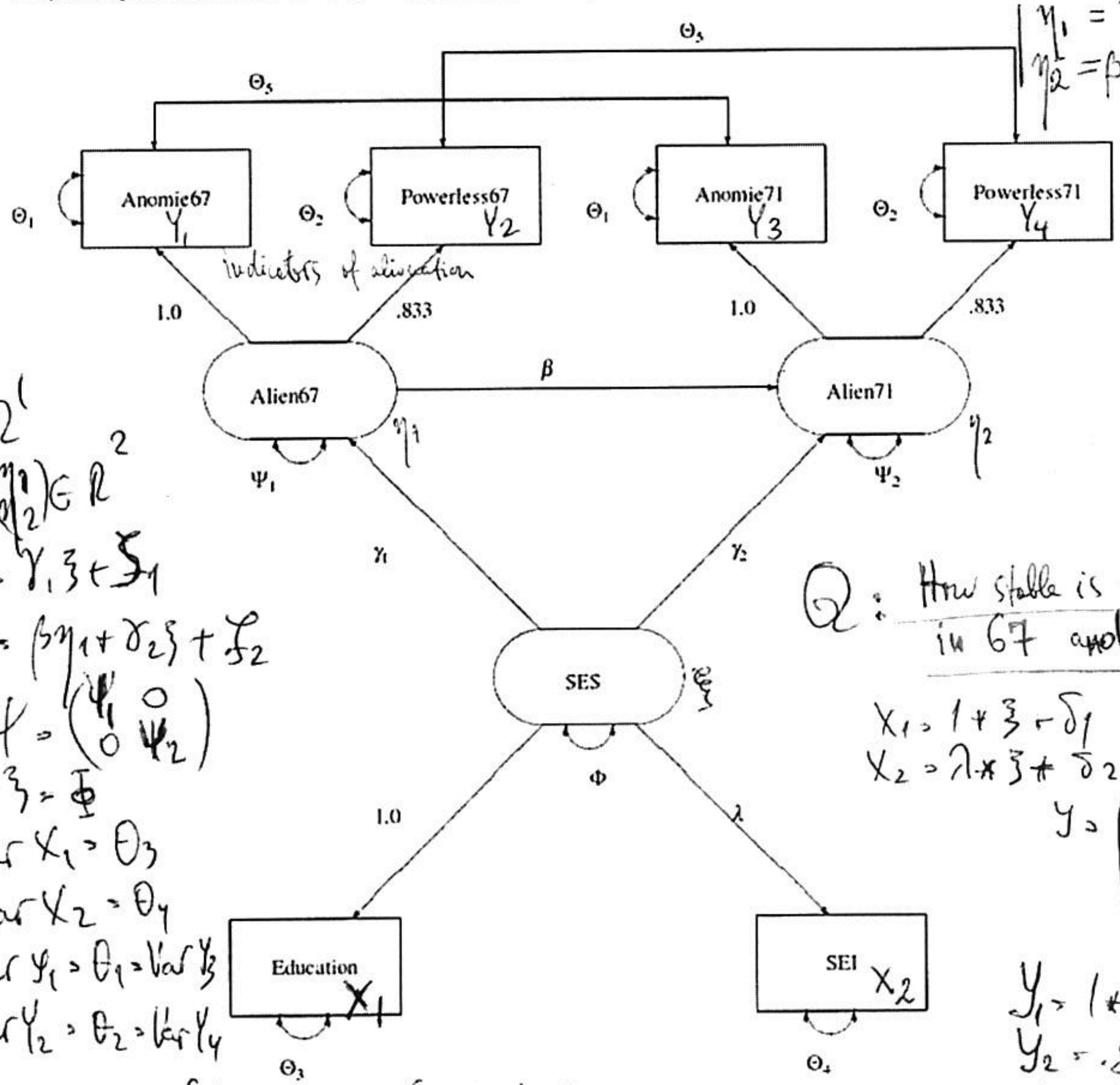
The CALIS Procedure

Example 26.16 Path Analysis: Stability of Alienation

The following covariance matrix from Wheaton et al. (1977) has served to illustrate the performance of several implementations for the analysis of structural equation models. Two different models have been analyzed by an early implementation of LISREL and are mentioned in Jöreskog (1978). You can also find a more detailed discussion of these models in the LISREL VI manual (Jöreskog and Sörbom; 1985). A slightly modified model for this covariance matrix is included in the EQS 2.0 manual (Bentler; 1985, p. 28). However, for the analysis with the EQS implementation, the SEI variable is rescaled by a factor of 0.1 to make the matrix less ill-conditioned. Since the Levenberg-Marquardt or Newton-Raphson optimization techniques are used with PROC CALIS, rescaling the data matrix is not necessary and, therefore, is not done here. The results reported here reflect the estimates based on the original covariance matrix.

The path diagram of this model is displayed in Figure 26.1 and is reproduced in the following:

SEM Model
$\eta_1 = \gamma_1 \xi + \zeta_1$
$\eta_2 = \beta \eta_1 + \gamma_2 \xi + \zeta_2$



$\xi \in \mathbb{R}^1$
$\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \in \mathbb{R}^2$
$\eta_1 = \gamma_1 \xi + \zeta_1$
$\eta_2 = \beta \eta_1 + \gamma_2 \xi + \zeta_2$
$\Psi = \begin{pmatrix} \Psi_1 & 0 \\ 0 & \Psi_2 \end{pmatrix}$
$Var\, \xi = \Phi$
$Var\, X_1 = \Theta_3$
$Var\, X_2 = \Theta_4$
$Var\, y_1 = \Theta_1 = Var\, y_3$
$Var\, Y_2 = \Theta_2 = Var\, Y_4$
$cov(y_2, y_4) = cov(y_1, y_3) = \Theta_5$

Q: How stable is alienation in 67 and in 71?

$X_1 = 1 * \xi + \delta_1$
$X_2 = \lambda * \xi + \delta_2$
$y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}$
$y_1 = 1 * \eta_1 + \varepsilon_1$
$y_2 = .833 * \eta_1 + \varepsilon_2$
$y_3 = 1 * \eta_2 + \varepsilon_3$
$y_4 = .833 * \eta_2 + \varepsilon_4$

You use the PATH modeling language of PROC CALIS to specify this path model, as shown in the following statements:

```
title "Stability of Alienation";
title2 "Data Matrix of WHEATON, MUTHEN, ALWIN & SUMMERS (1977)";
data Wheaton(TYPE=COV);
   _type_ = 'cov';
```